Non-Invasive Circadian Phase Assessment Using Wearable-Based Melatonin, Cortisol, and HRV Analysis

Summary

1. *Introduction*
   - 1.1 Circadian phase detection and physiological importance
   - 1.2 The need of a non-invasive solution
2. *Dataset preparation*
   - 2.1 Material review
   - 2.2 HRV feature selection
   - 2.3 Cortisol and Melatonin interpolation
3. *Melatonin Onset*
   - 3.1 Biological background
   - 3.2 Methodology
   - 3.3 Interpretation of the results
4. *Cortisol peak and Trough detection*
   - 4.1 Biological background
   - 4.2 Methodology
   - 4.3 Interpretation of the results
   - 4.4 Considerations on Nighttime Cortisol Peaks
5. *Feature Matrix and Label Vector*
   - 5.1 Objective
   - 5.2 Methodology
   - 5.3 Considerations

1. Introduction

1.1. Circadian phase detection and their physiological importance

Circadian rhythms are biological cycles that repeat roughly every 24 hours, driven by an internal timekeeping system often referred to as the "body clock." These rhythms influence a wide range of physiological processes, including sleep-wake patterns, hormone secretion, core body temperature, metabolism, and cognitive performance. Central to the regulation of circadian timing is the suprachiasmatic nucleus (SCN) in the hypothalamus, which responds to environmental cues like light and darkness to keep the internal clock in sync with the external world. In clinical and research settings, circadian phase—essentially, the timing of internal biological rhythms—is commonly assessed through the measurement of hormones such as cortisol and melatonin. Cortisol typically peaks in the early morning, as soon as we wake up, and decreases throughout the day, while melatonin levels rise in the evening, signalling the body to prepare for sleep. These markers are typically measured using repeated blood or salivary samples, often under controlled light conditions. However, while accurate, these methods are invasive, time-consuming, and not easily scalable for everyday monitoring. Understanding and detecting circadian phases is crucial for diagnosing sleep disorders, optimizing medication timing (chronotherapy), and managing conditions like depression, metabolic syndrome, and cancer, all of which can be influenced by disruptions in the circadian system.

1.2. The need of a non-invasive solution

Current approaches to determining circadian phase—primarily based on repeated salivary or blood sampling to measure melatonin and cortisol levels—are highly invasive, costly, and logistically challenging. These procedures require controlled environments, trained personnel, and repeated biological sampling across several hours and days, making them completely impractical for routine or real-life use. As a result, their application remains limited to specialized clinical or research settings, leaving a significant gap in accessible circadian monitoring for the general population. This underscores the urgent need for a non-invasive, low-burden alternative that can be integrated into everyday life. The development of wearable, sensor-based tools capable of estimating circadian rhythms passively and continuously would represent a crucial step toward personalized healthcare—enabling dynamic, real-time adaptation of treatments, lifestyle choices, and clinical decisions in harmony with an individual's internal biological timing.

Among various biosignals, Heart Rate Variability (HRV) is increasingly recognized as a promising physiological indicator for tracking circadian rhythms non-invasively. Indeed, HRV reflects autonomic nervous system activity, which itself exhibits circadian modulation, particularly through fluctuations in sympathetic and parasympathetic balance. An additional benefit of HRV is its non-invasive nature, as it can be accurately derived from ECG recordings obtained via wearable technology.

Building upon this foundation, the aim of this project is to design and implement a software tool capable of detecting circadian phase transitions from HRV features alone, in a completely non-invasive application. The core of this system relies on the development of a trained-and-tested AI model, able to correlate HRV values to cortisol and melatonin onsets, detecting in this way the patient's current circadian phase. By leveraging existing physiological insights and expanding them with computational techniques, the software aspires to provide a user-friendly, potentially GUI-integrated platform that could be used in every-day life conditions to detect internal biological time, without the need for hormonal testing or controlled laboratory settings.

2. Dataset preparation

2.1. Material review

The provided material consisted in:

- 10 datasets containing HRV features
- Hormonal reference datasets for Cortisol and Melatonin

The 10 HRV datasets used in this study were collected from 10 healthy subjects who wore wearable sensors over a period of two, or more, consecutive days. The primary signal of interest was the electrocardiogram (ECG), acquired continuously via a chest-worn device. From the raw ECG data, Heart Rate Variability (HRV) features were extracted by segmenting the signal into 5-minute non-overlapping windows. For each window, a comprehensive set of features was computed, covering time-domain metrics (e.g., mean HR, RMSSD, NNxx), frequency-domain parameters (using both Welch and Autoregressive methods), and non-linear dynamics (e.g., Poincaré descriptors, entropy measures, and recurrence analysis). Each row in the dataset corresponds to a single 5-minute window and includes timestamped information, allowing synchronization with other physiological markers (e.g. melatonin and cortisol onset). These features serve as the basis for circadian phase analysis, enabling the model to learn temporal patterns that are potentially indicative of internal biological timing.

To support the estimation of circadian phase, two complementary datasets were used containing time-stamped measurements of **salivary cortisol** and **melatonin** concentrations for 10 subjects across two consecutive days. These hormonal biomarkers serve as gold standards for identifying the circadian cycle's key reference points.

- **Cortisol dataset**: For each subject, cortisol levels were measured at regular intervals throughout the day on both Day 1 and Day 2. Each row includes the volunteer ID, the exact date and time of the sample, and the corresponding cortisol value.

- **Melatonin Dataset**: Similarly to the cortisol file, each row includes the subject ID, the sample collection time, and the melatonin concentration. The data allow for the detection of melatonin onset (MO), which generally occurs in the evening and is considered a reliable marker of internal circadian phase. The dataset includes separate entries for Day 1 and Day 2 for each subject, enabling the analysis of daily variability and supporting validation of HRV-based circadian phase estimation.

2.2. Cortisol and Melatonin interpolation

In order to align hormonal data with high-resolution HRV measurements, interpolation of melatonin and cortisol values was necessary. Cortisol and melatonin concentrations were obtained from discrete saliva samples collected at irregular intervals across two consecutive days, typically every 2 to 4 hours. In contrast, HRV features were computed over uniform 5-minute segments, resulting in a much finer temporal resolution. To enable accurate temporal correlation between HRV features and hormonal profiles, the sparse hormonal measurements had to be transformed into continuous time series.

Cortisol and Melatonin interpolation was achieved in MATLAB, covering the following steps:

1. Data loading: the Cortisol and Melatonin data, along with the HRV file for the selected patient were loaded using the *readtable* function.

2. Time parsing: the time samples contained in the HRV files were converted into MATLAB *datetime* format. Then, the mid-point of each 5 minutes interval was extracted. The *MidTime* value is crucial for aligning HRV features with interpolated hormone levels and for building accurate time-series models where each data point needs a precise timestamp.
3. Cortisol data extraction: from the Cortisol data file were extracted the columns corresponding to the patient's id, the date and time indication, and the cortisol value for both day 1 and day 2 (for a total of six columns, three per day). The columns were then filtered to maintain exclusively the information related to the selected patient. Finally, date and time information was parsed, and the cortisol levels were saved into a variable.
4. Melatonin data extraction: the same procedure applied for cortisol was repeated for handling melatonin data.
5. Sorting Cortisol and Melatonin data: the *sort* function was applied to ensure hormone levels were reported in a strictly chronological order, which is a fundamental requirement to perform correct data interpolation.
6. Interpolation: To align hormone concentrations with the higher-resolution HRV data, linear interpolation was applied using MATLAB's *interp1* function. This allowed the estimation of cortisol and melatonin values at the midpoint of each 5-minute HRV segment. The 'linear' method was chosen for its simplicity and suitability for sparsely sampled hormonal data, providing a smooth and realistic approximation of hormone trends between measurement points. The 'extrap' option was included to ensure that interpolated values could be computed even for time points falling slightly outside the original sampling window, thus avoiding gaps in the aligned dataset.
7. Creation of a merged CSV file
8. Visual plot of the interpolated data

The output of this MATLAB pipeline is a merged CSV file containing the patient's HRV features with three added columns: the first displaying the midpoint of each 5 minutes interval, the second and third columns reporting respectively the Cortisol and Melatonin interpolated levels.

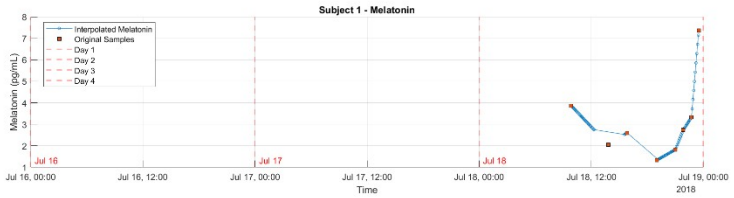
N.B.: The developed code handled one patient at a time, hence it was ran 10 times, producing 10 different merged csv files, one for each subject.

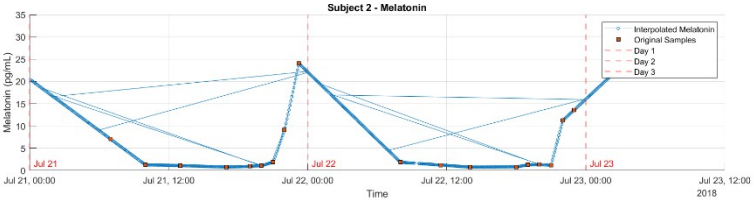
Particular attention was required in the processing of Subject 1, as melatonin data from Day 1 was missing. To address this, the melatonin series was first filtered to retain only valid measurements from Day 2, and interpolation was then performed exclusively on this subset. This approach ensured a smoother and more reliable time series, avoiding the introduction of artifacts due to the absence of early data. The lack of Day 1 values is clearly visible in both the time-series visualization and the corresponding merged CSV file, where melatonin entries are absent for that day.

Subject 1 Merged

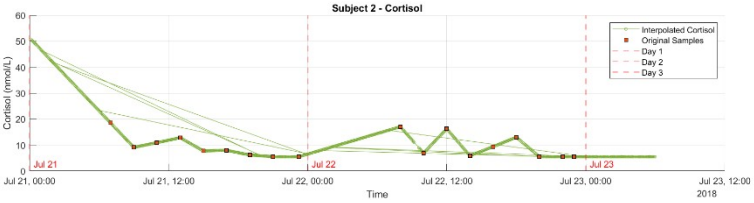
	Determinism	Shannon_Entropy	MidTime	Cortisol	Melatonin
1	0.994683607	3.715847743	16-Jul-2018 22:53:27	23.0332727272727	NaN
2	0.977069546	3.301207948	16-Jul-2018 22:57:58	22.984	NaN
3	0.984538075	3.446433189	16-Jul-2018 23:02:30	22.9345454545455	NaN
4	0.972117558	2.980937016	16-Jul-2018 23:07:31	22.8798181818182	NaN
5	0.973362649	3.096194168	16-Jul-2018 23:12:32	22.8250909090909	NaN
6	0.987831207	3.390787348	16-Jul-2018 23:17:33	22.7703636363636	NaN
7	0.964952107	2.982604613	16-Jul-2018 23:22:34	22.7156363636364	NaN
8	0.944376187	2.743829722	16-Jul-2018 23:27:35	22.6609090909091	NaN

When comparing the plots of Subject 1 and Subject 2, the difference becomes evident: Subject 1's melatonin profile shows a discontinuous curve limited to Day 2, while Subject 2's plot presents a fully interpolated and continuous melatonin curve spanning three consecutive days.









4. Melatonin Onset

4.1 Biological background

The Melatonin Onset (MO) refers to the first measurable rise in melatonin in the evening,

Typically occurring between 21:00 and 23:00, under normal circadian conditions. MO detection provides a direct, objective, and reliable marker of an individual's circadian phase, essential for diagnosing circadian rhythm sleep disorders, optimizing melatonin administration, and advancing personalized medicine.

The objective of this part of the study is to detect the Melatonin Onset (MO) for each subject based on interpolated melatonin concentration measurements and to label 5-min HRV windows as:

- 0: before MO
- 1: after MO

4.2 Methodology

The dataset used is the merged CSV file obtained from the previous step. In MATLAB, a melatonin onset detection pipeline was implemented based on the 3k method to label 5-min HRV windows for circadian phase analysis.

This method adapts to each subject's baseline melatonin level, instead of applying a fixed concentration threshold, making it more biologically accurate. The steps followed are:

1. Data loading: the merged CSV file was imported using the *readtable* function.
2. Chronological sorting: the dataset was sorted based on the center time of each 5-min window (*MidTime*) to ensure correct temporal ordering.

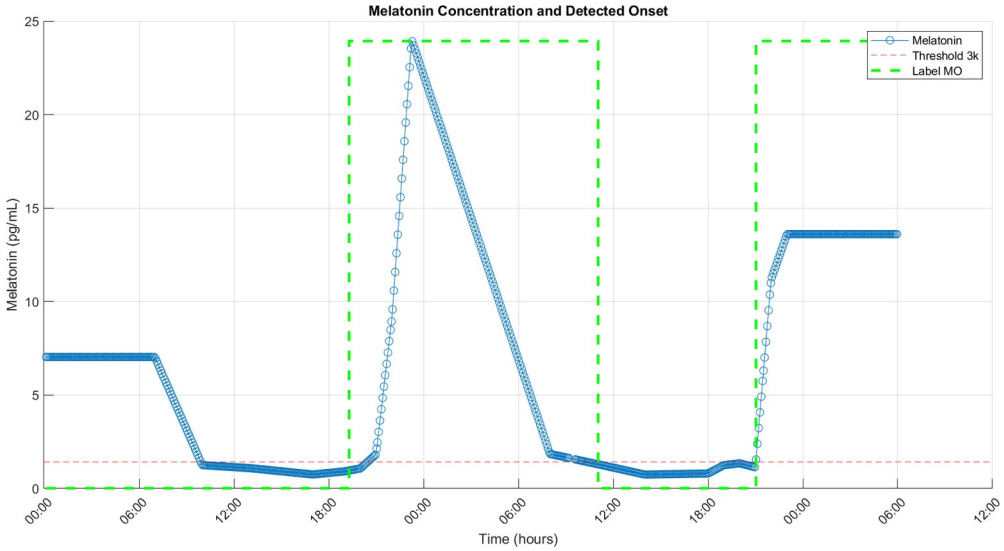
3. Definition of biological nights: Biological nights are defined automatically as periods from 18:00 of a given day to 11:00 of the following morning, to capture the natural transition from evening into early daytime. This structure ensures that melatonin onset, which generally occurs after sunset, is detected within the biologically relevant window.

4. MO detection (3k method): it is a very precise and robust way of detecting melatonin onset from salivary melatonin profiles. It is used especially when sampling is sparse (e.g., one sample/hour). It computes a baseline melatonin level from the lowest daytime melatonin values and defines onset as the first timepoint when melatonin exceeds baseline + 3× the standard deviation (k = 3).
   In this study, 12 daytime melatonin values were used instead of 6 to obtain a more stable and reliable estimate of the baseline. This adjustment reduces the risk of underestimating the threshold and detecting false early onsets caused by noise fluctuations. While early melatonin onset is possible in some subjects with an advanced circadian phase, this method ensures closer alignment with the expected biological timing in most cases. The detailed technical steps were:
   a. Sorting the dataset chrnologically
   b. Defining two biological night periods
   c. Compute baseline and threshold per night
   d. Lable each HRV window

4.3 Interpretation of the results

The main outcome is identification of the melatonin onset time for each subject. For better visualization of the results a melatonin concentration vs. time graph was plotted showing:

- The melatonin concentration (blue curve)
- The 3k threshold (red dashed line)
- The binary MO label as a green dashed step function



Melatonin onset of subject 2

Visual validation confirms that the detected MO aligns with major rises in melatonin concentration, even though slight deviations from expected timing may occur due to subject variability, baseline estimation noise or sampling precision.

Overall, the melatonin onset detection and labelling pipeline provides a robust and biologically meaningful classification of HRV segments relative to circadian phase.

5. Cortisol Peak and Trough Detection

5.1 Biological Background

Cortisol is a key circadian-regulated hormone that typically exhibits a high-amplitude daily rhythm, peaking in the early morning hours (around 06:00–09:00) and reaching its lowest concentrations during the nighttime.

The Cortisol Peak reflects the activation of the hypothalamic-pituitary-adrenal (HPA) axis and is essential for initiating the biological "wake-up" processes in the human body. Similarly, the Cortisol Trough represents the nadir of cortisol concentration, often occurring during the night.

Detecting cortisol peaks and troughs provides important information about the integrity and alignment of the circadian system, and can support clinical evaluation of different conditions.

5.2 Methodology

The dataset used is the same merged CSV file processed in the previous step. The following procedure was implemented in MATLAB to detect cortisol peaks and troughs for each subject

- Data Loading: the merged file was loaded using the *readtable* function and sorted chronologically based on *MidTime*.
- Definition of biological days: they were automatically detected by shifting all timestamps to their corresponding date (removing time of day information). Unique days were extracted. Biological days were defined to match the subject's cortisol sampling periods across two or more full calendar days.
- Peak and trough detection for each biological day separately:
    - The maximum cortisol value corresponds to the Cortisol Peak.
    - The minimum cortisol value corresponds to the Cortisol Trough.

  In cases where multiple time points shared the same maximum or minimum cortisol concentration, all matching time points were labeled to reflect physiological plateaus.
- Labeling: a new binary column *Label_CortPeak* was created:
    - 1 if the time point corresponds to a cortisol maximum.
    - 0 otherwise.

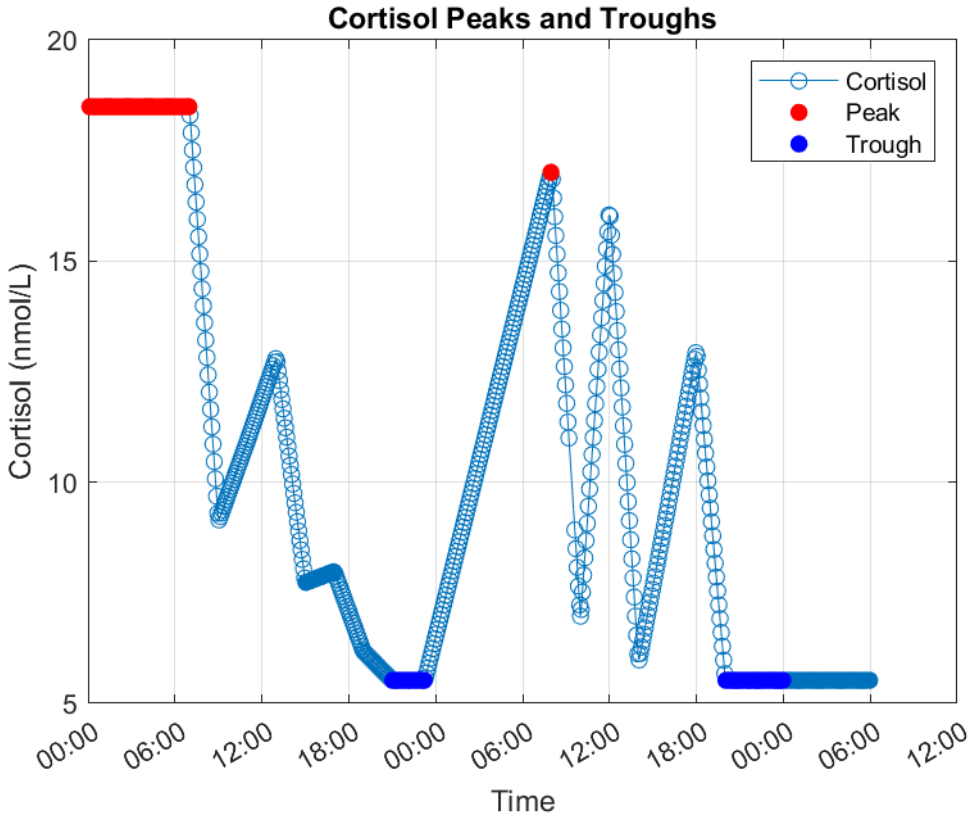
  A second binary column *Label_CortTrough* was created:
    - 1 if the time point corresponds to a cortisol minimum.
    - 0 otherwise.

5.3 Interpretation of the results

The primary output is the identification of the Cortisol Peak and Trough times for each biological day.

To visually validate the results, a cortisol concentration vs. time plot was generated, displaying:

- The cortisol profile (blue curve),
- Red markers for cortisol peaks,
- Blue markers for cortisol troughs.



Cortisol concentration for subject 2

The peak values were found to occur within expected physiological windows, typically in the early morning hours, while trough values occurred during nighttime, consistent with circadian regulation of cortisol secretion.

5.4 Considerations on Nighttime Cortisol Peaks

During the analysis of cortisol profiles, it was observed that in some subjects, the maximum cortisol concentration occurred during nighttime hours, rather than in the early morning as typically expected in a healthy circadian rhythm.

Several biological and experimental factors can explain this observation:

- Circadian misalignment: Some individuals may have delayed or irregular circadian rhythms, leading to abnormal timing of cortisol secretion (e.g., shift work, sleep disorders).
- Acute stress events: As cortisol is a stress hormone, nocturnal stress responses could cause temporary elevations.
- Sparse sampling or missing data: If early morning cortisol measurements are lacking, the true biological peak might not be captured, and an artificial nighttime peak could appear instead.
- Natural biological variability: Inter-individual differences, age, medication use, or lifestyle factors can influence cortisol timing.

In this study, the cortisol peak detection method faithfully labels the highest measured cortisol value within each day. Thus, nighttime peaks were preserved rather than artificially corrected, ensuring that the labeling remains data-driven and unbiased.

6. Feature Matrix and Label Vector

6.1 Objective

To train supervised machine learning models for melatonin onset detection and cortisol peak detection, it is necessary to organize the HRV data into:

- A feature matrix (X) containing physiological descriptors.
- A corresponding label vector (y) indicating the circadian phase or hormonal event.

This structure allows the models to learn the relationship between HRV patterns and biological timing.

6.2 Methodology

A consistent set of 19 HRV features was selected for both melatonin and cortisol analyses, covering time-domain, frequency-domain, and non-linear descriptors of heart rate variability.

The feature matrix X was built by extracting the selected HRV features. The label vector y was built by extracting the corresponding label (*Label_MO* or *Label_CortPeak*). Samples with missing labels were removed to ensure that only well-defined training examples remained. Both feature matrices were normalized using z-score normalization (mean = 0, standard deviation = 1) to improve machine learning model performance.

Each matrix was saved into .csv files named by subject, e.g., X_MO_subj_1.csv, y_CortPeak_subj_2.csv, etc., for organized data management.

6.3 Considerations

It is important to note that the set of HRV features is identical for both tasks, ensuring a consistent input space. However, the number of samples may differ between tasks: not all 5-minute HRV windows may be labeled the same for melatonin and cortisol events. As a result, X_MO and X_CortPeak may have different numbers of rows, although they share the same columns.

This separation reflects the biological differences between the timing of melatonin secretion onset and cortisol peak occurrence.